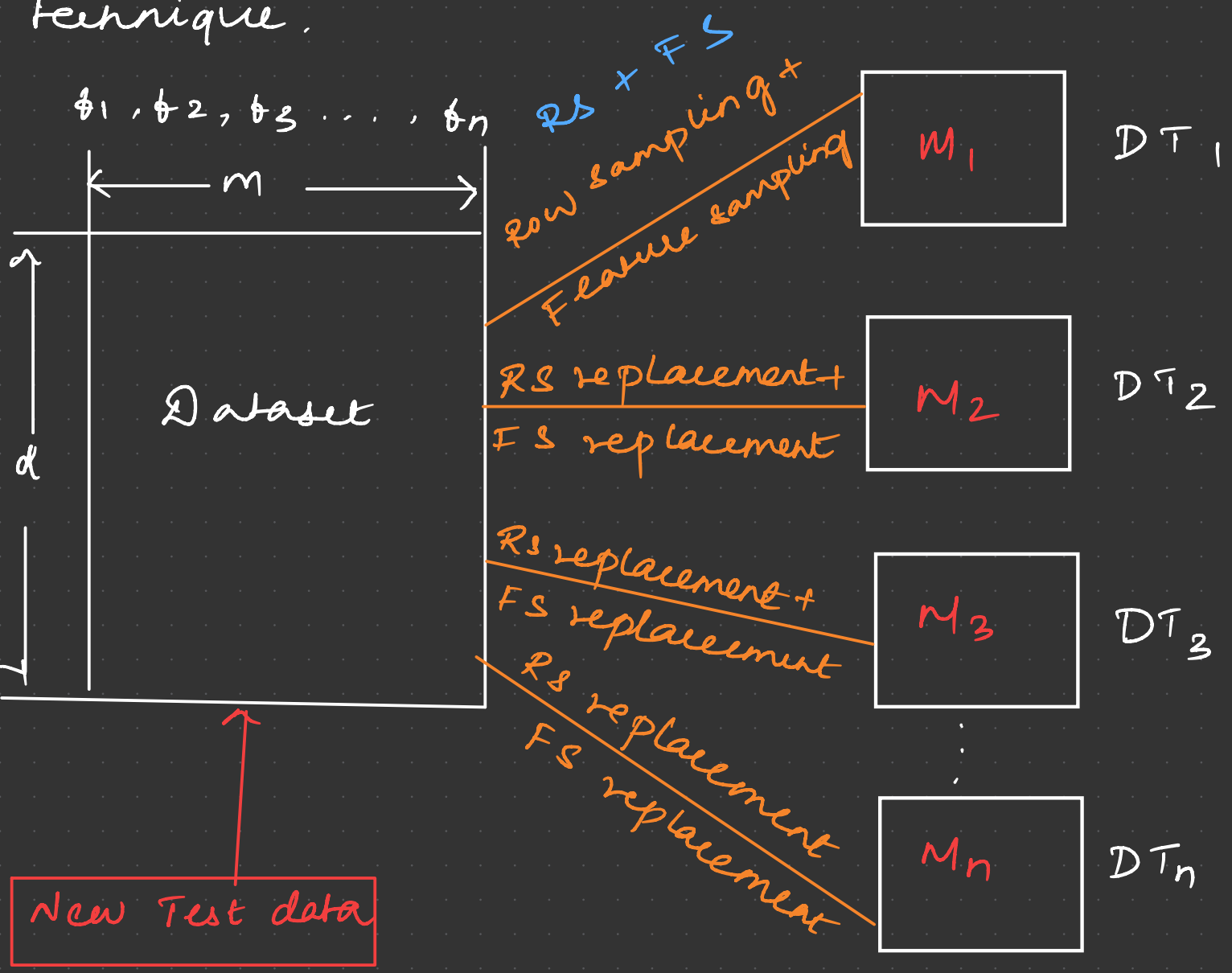
Random Forest Regression and classification

This is one of the bagging technique.



Classification
⇓
Majority voting classifier

Regression
⇓
Average of all model output.

Why should we use Random Forest Instead of decision tree

Decision tree

→ Overfitting → If training with the default parameters.

↳ Train accuracy ↑ → Low Bias
↳ Test accuracy ↓ → High variance

Generalized model → Low Bias
Generalized model → Low variance.

with the help of random forest we will make,

Low Bias → Low Bias
High variance → Low variance.

Idea here is to build the Generalized model.

Since, in random forest, we have multiple decision (multiple base learner) high variance would automatically converted to low variance.